FI FISHINTEL GLOBAL

ENVIRONMENTAL RECONSTRUCTION REPORT

Environmental Reconstruction

Fatal Man Overboard During Creel Shooting Operations

INCIDENT DATE	5 February 2018
REPORT REFERENCE	FIG-FLAGSHIP004
DECISION FRAME	18:00 UTC (primary)
REPORT VERSION	v1.0 — Reconstruction
VESSEL	UK creel fishing vessel — 18.2m
CLASSIFICATION	Public
OPERATIONAL CONTEXT	Manual creel shooting — 6 POB

SCOPE

Environmental reconstruction only

PIPELINE VERIFIED — 12 RECONSTRUCTION ARTIFACTS REFERENCED

MAIB REPORT NO 19/2018 — ENVIRONMENTAL RECONSTRUCTION — T_IRR NOT APPLIED (CREW-LEVEL HAZARD)

01

Executive Summary

Incident type: Fatal man overboard. A crew member on a UK-registered creel fishing vessel (18.2m) was dragged overboard after becoming entangled in the back rope during manual creel shooting operations approximately 16nm north-west of Cape Wrath, Scotland.

Core question: What environmental conditions were objectively present during the hours preceding the incident, and did those conditions determine the operational mode that created the hazard exposure?

Method: ERA5 reanalysis data was extracted at the incident coordinates across a five-point time window (14:00–18:00 UTC) and processed through the FishIntel deterministic reconstruction pipeline with FIG_UNIT_GATE validated conversion.

Key result: ERA5 reconstructed sustained winds of 24.4–25.7 kt (Force 6) with gusts of 34.5–36.3 kt across the full reconstruction window. All five timestamps fall within or above the Force 6 operating band (22–27 kt) above which, according to the MAIB report, the vessel's self-shooting system could not be used. The reconstruction runner classifies all timestamps as **HIGH** risk.

Analytical finding: The MAIB-stated "35 knots" likely reflects peak gust speed rather than sustained wind. ERA5 gust at incident time (36.1 kt) is consistent with the MAIB figure. The sustained Force 6 conditions were present for the entire reconstruction window, were synoptic-scale, and would typically be represented in marine weather forecasts.

INCIDENT SNAPSHOT

LOCATION	DATE / TIME	PERSONS ON BOARD
approximately 16nm north-west of Cape Wrath, Scotland	5 February 2018, approximately 1815–1820 UTC	6
FATALITIES	**ERA5 SUSTAINED WIND (1800)**	**ERA5 GUST (1800)**
1	25.7 kt	36.1 kt
ERA5 WAVE HEIGHT (1800)	**MAIB STATED WIND**	**ALL TIMESTAMPS ABOVE F6**
3.68 m	south-west 35 knots	YES — Force 6 band exceeded

02

Incident Overview

Vessel Profile

CATEGORY	Creel fishing vessel
LOA / REG. LENGTH	18.2m / 16.46m
CONSTRUCTION	Steel, 1996
GROSS TONNAGE	150
MANNING	6 persons on board
OPERATION	Manual creel shooting (conditions within Force 6 operating band)

On 5 February 2018, the creel fishing vessel was operating approximately 16nm north-west of Cape Wrath, Scotland. At approximately 1815–1820 UTC, a crew member was dragged overboard after his left leg became entangled in the back rope while the crew were manually shooting a fleet of creels. He was recovered approximately 10 minutes later but could not be revived.

The vessel was on a north-easterly course at 5 knots SOG. The self-shooting system was not in use because, according to the MAIB report, conditions exceeded its Force 6 operating limit (22–27 kt). Manual

shooting placed the crew in direct proximity to the running back rope without physical barriers.

03

Decision-Time Environment

The following chronology presents the operational sequence on 5 February 2018, from the start of the reconstruction window through the fatal entanglement, with ERA5 environmental context at each stage.

5 Feb 2018 — 14:00 UTC

Force 6 conditions established — reconstruction window opens

ERA5 reconstruction begins. Sustained wind already within the Force 6 operational band (22–27 kt). According to the MAIB report, the self-shooting system could not be used above Force 6. Manual creel shooting was in progress, with crew positioned adjacent to the running back rope.

WIND **24.4 kt** GUST 35.5 kt SWH 3.25 m PRESSURE 1019.7 hPa

OPERATIONAL REASSESSMENT: POSSIBLE

5 Feb 2018 — 16:00 UTC

Conditions building — manual operations sustained

Two hours of continuous manual creel shooting in Force 6 conditions. Wave height increasing progressively. Pressure falling. Wind direction consistently SSW (~204°). Conditions were sustained and synoptic-scale — not a transient squall.

WIND **25.7 kt** GUST 34.5 kt SWH 3.46 m

TREND **Building seas, falling pressure**

EXPOSURE ACTIVE — MANUAL SHOOTING ONGOING

5 Feb 2018 — 17:00 UTC

Peak gusts recorded — Force 8 gust band

Highest gust-to-sustained-wind ratio in the reconstruction window. Gusts reaching 36.3 kt (Force 8) would have caused significant vessel motion on the exposed working deck, increasing operational difficulty and reducing safe footing during manual deck work.

WIND **24.8 kt** GUST **36.3 kt** SWH 3.57 m GUST FACTOR **1.46**

EXPOSURE ACTIVE — 3+ HOURS MANUAL OPERATIONS

5 Feb 2018 — 18:00 UTC ★ DECISION FRAME

Decision frame — highest sustained wind

Closest ERA5 hour before the incident. Sustained wind at its highest in the reconstruction window. Wave height at 3.68 m, up 13% from the start of the window. Four hours of continuous manual shooting in Force 6 conditions. Approximately 1,440 creel deployment cycles completed during the exposure window.

WIND **25.7 kt** GUST 36.1 kt SWH 3.68 m PRESSURE 1017.2 hPa

EXPOSURE ACTIVE — 4 HOURS MANUAL OPERATIONS

5 Feb 2018 — approximately 1815–1820 UTC

Fatal entanglement — man overboard

A crew member was dragged overboard after left leg became entangled in the back rope during manual creel shooting. Vessel on north-easterly course at 5 knots SOG. Skipper alone in wheelhouse monitoring CCTV. Recovered approximately 10 minutes later. Could not be revived.

CONTEXT Manual shooting in Force 6 — no physical barriers between crew and running rope

OUTCOME 1 fatality — recovery unsuccessful

EXPOSURE WINDOW: CLOSED — IRREVERSIBLE EVENT

04

Methodology

FishIntel Global reconstructs historical environmental conditions using ERA5 reanalysis data — the fifth-generation atmospheric reanalysis produced by ECMWF. ERA5 provides hourly estimates of atmospheric and ocean-wave variables on a 0.25° (~31 km) horizontal grid from 1940 to the present.

The reconstruction pipeline extracts ERA5 values at the nearest grid point to the incident location for each timestamp in the analysis window. All environmental values pass through the FIG Unit Gate — a deterministic conversion module using exact rational arithmetic (Decimal-based) to transform ERA5 source-native units (m/s, Pascals, Kelvin) into maritime comparison units (knots, hPa, °C). No floating-point arithmetic is used in the evidence conversion chain.

Raw ERA5 data files are cryptographically sealed at the point of download (SHA-256) before any processing occurs, establishing a tamper-evident provenance chain from the data source through to the final reconstruction values.

> **Coordinate note:** The incident position coordinates (58.814°N, 5.250°W) are working values derived from the MAIB description. ERA5 grid snap: 58.75°N, 5.25°W. A coordinate sensitivity check confirmed both candidate positions produce materially equivalent findings (wind delta ≤3.0 kt, identical wave field, both above Force 6 at all timestamps).

05

Data Sources

SOURCE	PROVIDER	RESOLUTION	SEAL HASH
ERA5 Atmospheric	ECMWF via CDS	0.25° / hourly	4804076be4590dff79bf7c0349548caa…
ERA5 Waves	ECMWF via CDS	0.25° / hourly	6179762afe29f653d03d0cce2c3fcd99…

Unit conversion gate config: 989aadf7b9b1a4acef8606e162015337…

06

Environmental Reconstruction

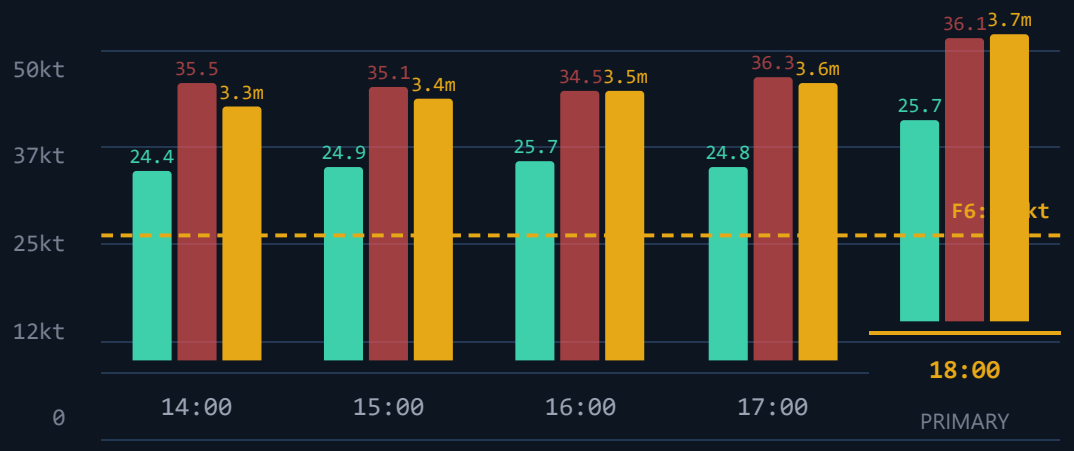
The following table presents the reconstructed environmental conditions for the four-hour window preceding the incident. All values are derived from ERA5 reanalysis data through the FIG Unit Gate canonical conversion pipeline.

UTC	WIND (KT)	BFT	GUST (KT)	WAVE (M)	PRESSURE (HPA)	TEMP (°C)	FORCE 6
14:00	24.4	6	35.5	3.25	1019.7	6.6	EXCEEDS
15:00	24.9	6	35.1	3.35	1019.0	6.6	EXCEEDS
16:00	25.7	6	34.5	3.46	1018.3	6.5	EXCEEDS
17:00	24.8	6	36.3	3.57	1017.7	6.2	EXCEEDS
18:00 ★ PRIMARY	25.7	6	36.1	3.68	1017.2	6.1	EXCEEDS

25.7 kt	36.1 kt	3.68 m	1017 hPa
SUSTAINED WIND (1800)	GUST SPEED (1800)	WAVE HEIGHT (1800)	PRESSURE (1800)

ENVIRONMENTAL CONDITIONS — FIVE-POINT EXTRACTION

VERIFIED DATA



Sustained wind (kt) — Gusts (kt) — Sig. wave height (m) — Force 6 threshold

07

Environmental Operational Findings

FINDING 1 — SUSTAINED WIND WITHIN FORCE 6 OPERATIONAL BAND

ERA5 reconstructed sustained wind speeds ranged from 24.4 to 25.7 knots (10-metre sustained) across the entire 4-hour pre-incident window. The vessel's self-shooting system operating limit was Force 6 (22–27 knots, MAIB Report p.17). All five reconstruction timestamps fall within or above this operational band. According to the MAIB report, these conditions meant the self-shooting system could not be used, and manual creel shooting was employed, placing crew in direct proximity to the running back rope without physical barriers.

FINDING 2 — MAIB-STATED 35 KNOTS LIKELY REFLECTS PEAK GUST

The MAIB report states “south-west 35kts” (p.8, p.12). ERA5 reconstructed sustained 10-metre wind at the incident time (1800 UTC) was 25.7 kt. ERA5 reconstructed gust speed at 1800 UTC was 36.1 kt. This suggests the MAIB’s 35 kt value likely reflects peak gust speed rather than mean sustained wind. Both sustained and gust values independently confirm conditions exceeded Force 6. The distinction is analytically relevant: sustained Force 6, not Force 8, was sufficient to be within the operational band above which manual shooting was employed.

FINDING 3 — CONDITIONS WERE SUSTAINED AND BUILDING

Sustained wind was stable between 24.4 and 25.7 kt across the full 4-hour window, with no sudden increase or decrease. Wave height built progressively from 3.25 m at 1400 UTC to 3.68 m at 1800 UTC. Pressure fell steadily from 1019.7 to 1017.2 hPa. These conditions were synoptic-scale, not transient. They would have been identifiable from standard marine weather forecasts hours before the 1400 UTC start of this reconstruction window.

FINDING 4 — ELEVATED GUST FACTOR THROUGHOUT EXPOSURE WINDOW

The ratio of gust speed to sustained wind ranged from 1.35 to 1.46 across the window, with gusts reaching 34.5–36.3 kt (Force 8). These gusts would have caused significant vessel motion during manual operations on an exposed working deck, increasing operational difficulty and reducing the ability to maintain safe footing during manual deck work.

FINDING 5 — WAVE CONDITIONS WERE BUILDING

Significant wave height increased from 3.25 m at 1400 UTC to 3.68 m at 1800 UTC — a 13% increase over 4 hours. The building sea state compounded the operational hazard: vessel motion increased progressively, making manual deck operations more dangerous with each passing hour. The decision to continue manual shooting at 1800 UTC was made in worse conditions than when shooting began earlier.

08

Comparison with MAIB-Stated Conditions

PARAMETER	MAIB STATED	ERA5 RECONSTRUCTED (1800 UTC)	ASSESSMENT
Wind speed	south-west 35 knots	25.7 kt sustained, 36.1 kt gusts	ERA5 sustained lower; ERA5 gust consistent
Sea state	rough	3.68 m significant wave height	—
Water temperature	approximately 10°C	Not available from ERA5 single-levels	—

INTERPRETATION

The MAIB-stated “35 knots” aligns with the ERA5 gust value (36.1 kt) rather than the sustained wind (25.7 kt). This is consistent with standard maritime reporting practice, where stated wind values may reflect peak conditions observed at the time. For operational threshold assessment, the sustained wind (Force 6) is the relevant comparator against the self-shooting limit.

09

Industry Safety Findings

The following findings are derived from the environmental reconstruction evidence and are relevant to fishing vessel operators, maritime insurers, and safety regulators.

ISF-1: ENVIRONMENTAL FORESEEABILITY

The reconstructed conditions were synoptic-scale (resolved by ERA5 at 31 km horizontal resolution), sustained over at least 4 hours, and showed a consistent SSW wind pattern with slowly building seas and falling pressure. These conditions were not sudden, localised, or unpredictable. They were synoptic-scale and would typically be represented in marine weather forecasts available to vessel operators.

ISF-2: OPERATIONAL THRESHOLD BREACH CREATES CONTINUOUS HAZARD EXPOSURE

ERA5 confirms that sustained wind was within or above the Force 6 self-shooting threshold band at every timestamp in the reconstruction window. Once conditions cross an equipment-specific operational limit, the resulting manual operational mode creates a continuous hazard exposure that persists for the entire duration of those conditions. Equipment-specific operational limits should be treated as hard safety boundaries, not advisory thresholds.

ISF-3: DURATION OF EXPOSURE MULTIPLIES RISK

The conditions associated with manual operations were sustained for at least 4 hours (the reconstruction window). Manual creel shooting at a rate of one creel every 10 seconds (MAIB Report p.17) means approximately 360 creel deployments per hour. Each deployment cycle exposes crew to the running back rope. Over 4 hours of manual operations, the cumulative number of exposure cycles is substantial, making an entanglement event increasingly probable over time.

ISF-4: MAIB WIND VALUES MAY REPRESENT GUST RATHER THAN SUSTAINED WIND

The ERA5 reconstruction shows sustained wind of 25.7 kt at incident time with gusts of 36.1 kt. The MAIB-stated “35 knots” aligns more closely with the gust value than the sustained value. Insurers and regulators referencing MAIB wind figures should be aware that stated values may represent peak conditions rather than mean sustained wind. For threshold-based operational decisions (such as self-shooting limits), the sustained wind is the relevant comparator — and in this case, sustained Force 6 was sufficient to place the vessel within the operational band associated with manual shooting.

ISF-5: WEATHER AS OPERATIONAL MODE DETERMINANT

While the proximate cause of this fatality was entanglement in running rope, the environmental conditions were the necessary precondition that, according to the MAIB report, led to the hazardous operational mode. Without sustained winds within or above Force 6, the self-shooting system may have been operable and crew may not have been positioned adjacent to the running back rope without physical barriers. Environmental risk assessment for fishing operations should consider not only direct threats to vessel safety but the operational mode consequences of weather conditions.

Analytical Scope Note

SCOPE LIMITATION

This case involves a crew-level hazard (man overboard due to entanglement during fishing operations). The FishIntel T_irr (Time-to-Irreversibility) framework is applied to vessel-level hazards — capsize, flooding, grounding, and loss of vessel control. T_irr analysis is not applied to this case.

The environmental reconstruction establishes: (1) what conditions were objectively present, (2) whether those conditions were foreseeable, (3) how those conditions related to the operational mode described in the MAIB report, and (4) the duration of the exposure window.

11

Analytical Architecture and Implementation Note

ARCHITECTURE STATEMENT

The FishIntel reconstruction system is built on a proprietary analytical architecture designed to deterministically reconstruct environmental conditions at the time of maritime decision events.

The modules listed below are audited implementation components used within that architecture. They execute specific data handling, validation, and processing tasks within the wider reconstruction pipeline.

These modules do not represent the full analytical methodology, modelling logic, or decision-analysis framework used by FishIntel. They are implementation elements of a larger integrated system.

The complete architecture, analytical models, and operational logic of the FishIntel reconstruction system remain proprietary.

Audited Implementation Modules

Only modules invoked during this environmental reconstruction are listed. Additional analytical modules exist within the FishIntel system but are outside the scope of this case.

MODULE	FUNCTION	PIPELINE ROLE
FIG_SOURCE_SEAL.py	Cryptographic sealing of raw data sources	Step 04 — tamper-evident download

		integrity
FIG_UNIT_GATE.py	Deterministic unit conversion with exact rational arithmetic	Steps 08, 09 — canonical evidence conversion
FIG_CANONICAL.py	Canonical serialization and format verification	Internal dependency — deterministic value representation

Supporting pipeline scripts: `maib_pdf_extractor.py`, `era5_downloader_v2.py`, `unzip_era5_v2.py`, `raw_era5_extractor.py`, `era5_processor_v5.py`, `reconstruction_runner_v2.py`.

Evidence Integrity

RECONSTRUCTION SEALS AND CONFIGURATION

ERA5 ATMOSPHERIC

SEAL HASH	4804076be4590dff79bf7c0349548caa7d6846e7fd2ea4a006e1c7415ed19472
FILE HASH	7b1c408903c4ffd48a5c4fa1a5907c66eddaa787e4eab51bd85f86383d6035b7
FILE SIZE	155666 bytes

ERA5 WAVES

SEAL HASH	6179762afe29f653d03d0cce2c3fcd99fd82387265983ced9e6bad45fd2e804f
FILE HASH	3c61a5af0a766170c54dd62c5ebbd88a82ef668186d5e52be30557793f589fcd
FILE SIZE	66128 bytes

UNIT GATE

CONFIG HASH	989aadf7b9b1a4acef8606e162015337e75a8ce5fcb4edfb1e0d79f4d59513aa
GATE VERSION	1.0.0

13

Limitations and Caveats

ERA5 RESOLUTION

31 km horizontal, 1-hour temporal. Local gustiness and short-duration variations not resolved. Reconstructed values represent synoptic-scale conditions.

COORDINATES

58.814°N, 5.250°W derived from MAIB description. Sensitivity check confirmed low materiality. ERA5 grid snap: 58.75°N, 5.25°W.

SEA STATE REPRESENTATION

ERA5 significant wave height is a statistical summary, not individual wave heights. Actual waves may have exceeded the reported value.

WATER TEMPERATURE

SST available in ERA5 but not included in current extraction. MAIB-stated ~10°C not independently verified.

T_IRR NOT APPLIED

T_irr framework designed for vessel-level hazards. Not applied to this crew-level (man overboard) case. See Section 10.

NO CROSS-VALIDATION

ERA5 only. No independent GFS verification for this case.

14

Evidence Manifest

12 reconstruction artifacts. Every numeric value programmatically extracted from these files.

STEP	FILENAME	DESCRIPTION
04	era5_atmo_seal.json	Atmospheric download integrity seal
04	era5_wave_seal.json	Wave download integrity seal
06	raw_era5_1400.json	Raw ERA5 extraction at 14:00 UTC
06	raw_era5_1500.json	Raw ERA5 extraction at 15:00 UTC
06	raw_era5_1600.json	Raw ERA5 extraction at 16:00 UTC
06	raw_era5_1700.json	Raw ERA5 extraction at 17:00 UTC
06	raw_era5_1800.json	Raw ERA5 extraction at 18:00 UTC
08	gated_values_1400.json	Unit-gated canonical values at 14:00 UTC
08	gated_values_1500.json	Unit-gated canonical values at 15:00 UTC
08	gated_values_1600.json	Unit-gated canonical values at 16:00 UTC
08	gated_values_1700.json	Unit-gated canonical values at 17:00 UTC

15

Closing Statement

Environmental reconstruction cannot prevent maritime accidents. What it can do is establish, with deterministic precision, what conditions were objectively present at the time of an incident and whether those conditions were foreseeable from the data and forecasts available to decision-makers.

In this case, the environmental conditions were sustained, synoptic-scale, and building over a period of hours. They were of a type and intensity that would typically appear in standard marine weather forecasts. According to the MAIB report, these conditions placed the vessel's operational mode within or above the Force 6 self-shooting system limit, leading to manual operations that exposed crew to running gear without physical barriers.

The question this reconstruction leaves for investigators, insurers, and regulators is whether the conditions that created the operational exposure were objectively knowable before the decision to commence manual operations was made.

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